

principles of quantum mechanics. These simulations are considered to be comparatively difficult and drained immense computational resources. Despite this, a lot of research interests exist in this area. Scientists believe that the tools developed on *ab-initio* principles can assist in various ways that can significantly improve various aspects of human life. The tool could make notable contributions, for instance, in the designing of new drugs, new materials and in the development of new information-processing technologies such as quantum computing devices.

nano-archimedes is based on the many-body Wigner equation, a sophisticated (and yet, very intuitive) formalism that allows *ab-initio* (Monte Carlo) simulations in phase-space, even for strongly-correlated systems.

Wigner formalism

Wigner formalism is a time-dependent, full quantum, multi-dimensional model based on the concept of a quasi-distribution function defined in phase-space. The approach is very intuitive and totally equivalent to the well-known Schrödinger model. In this model, an invertible Wigner-Weyl transform exists, which converts wave functions into quasi-distribution functions, and vice-versa. From this angle the situation is not any different from classical mechanics, where different formalisms like Newtonian, Lagrangian and Hamiltonian exist, and can be more or less convenient, depending on the system under consideration.

A system in Wigner formalism is described in terms of a quasi-distribution function defined in the phase-space of n -particles. Hence, we can consider this formalism as a very intuitive approach that is closer to the way experimentalists perform their experiments. Wigner equation allows simulation of many-body quantum systems in a time-dependent, multi-dimensional fashion. This allows scientists to simulate ground and excited states.



nano-archimedes is a very exciting project. There is no doubt that the next-generation technology will be strongly based on our comprehension of quantum mechanics. Our software is the first complete and working, full-quantum, multi-dimensional and time-dependent simulator released under GPL. This means that anyone can use it, modify it and share it. Despite being released only recently, I can already see this code being perceived by the community as a tool on which to construct even more advanced results. I am sure that it represents a concrete advance towards the comprehension and design of practical quantum devices."

— Jean Michel Sellier, main developer of nano-archimedes

The iterative Monte Carlo method—a general time-dependent approach for partial differential equations that can deal with general initial and boundary conditions—is utilised to numerically solve Wigner equation. Monte Carlo techniques scale incredibly well on parallel machines, allowing the simulation of comparatively very complex systems. We have discussed about Monte Carlo simulations in detail in March issue of EFY.

The Wigner formalism is nowadays applied to a plethora of different situations. For example, recently it has been utilised in the field of simulation of semiconductor devices, post-CMOS design and quantum chemistry (Wigner DFT and many-body techniques), among others. nano-archimedes software is based on this Wigner formalism and could be used in advanced research in these areas.

Signed particle formulation to be introduced into nano-archimedes

The latest development in nano-archimedes is based on a brand new formulation of quantum mechanics called signed particle formulation, which is currently under review. This is an important generalisation of Wigner MC method provided with a physical interpretation. In this novel formulation, a quantum system is described in terms of virtual Newtonian field-less particles that can interact with a potential by means of creation and annihilation of particles only.

Being a very intuitive approach, thus easy to implement, the signed particle formulation proves to be more convenient than other approaches in certain cases of quantum system simulations. "Even though the signed particle formulation is mathematically equivalent to other formulations, such as Schrödinger and Wigner, it has strong advantages to be intuitive, highly parallelisable and easy to extend to situations such as inelastic scattering," says J.M. Sellier.

Boosting up scientific innovations

In the initial days of research, there are times when we do not know where to start the work. If we consider the semiconductor industry, researches around the world have been so advanced that we are now dealing in nano-scale sizes. Unfortunately, the people who are new to this field of research have to begin their work from scratch, as every work will be patented and not available freely. This will waste a lot of time, effort and money.

Providing the source codes of Archimedes and nano-archimedes as open source is a positive step in this regard, as more researchers could benefit from it and could spend their time more fruitfully. Let us hope this step from the developers will inspire other researchers to release more free scientific codes, and thereby enhance more scientific advancements in the society. ●

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